1. Description

1.1. Project

Project Name	final
Board Name	NUCLEO-G474RE
Generated with:	STM32CubeMX 6.16.1
Date	03/11/2026

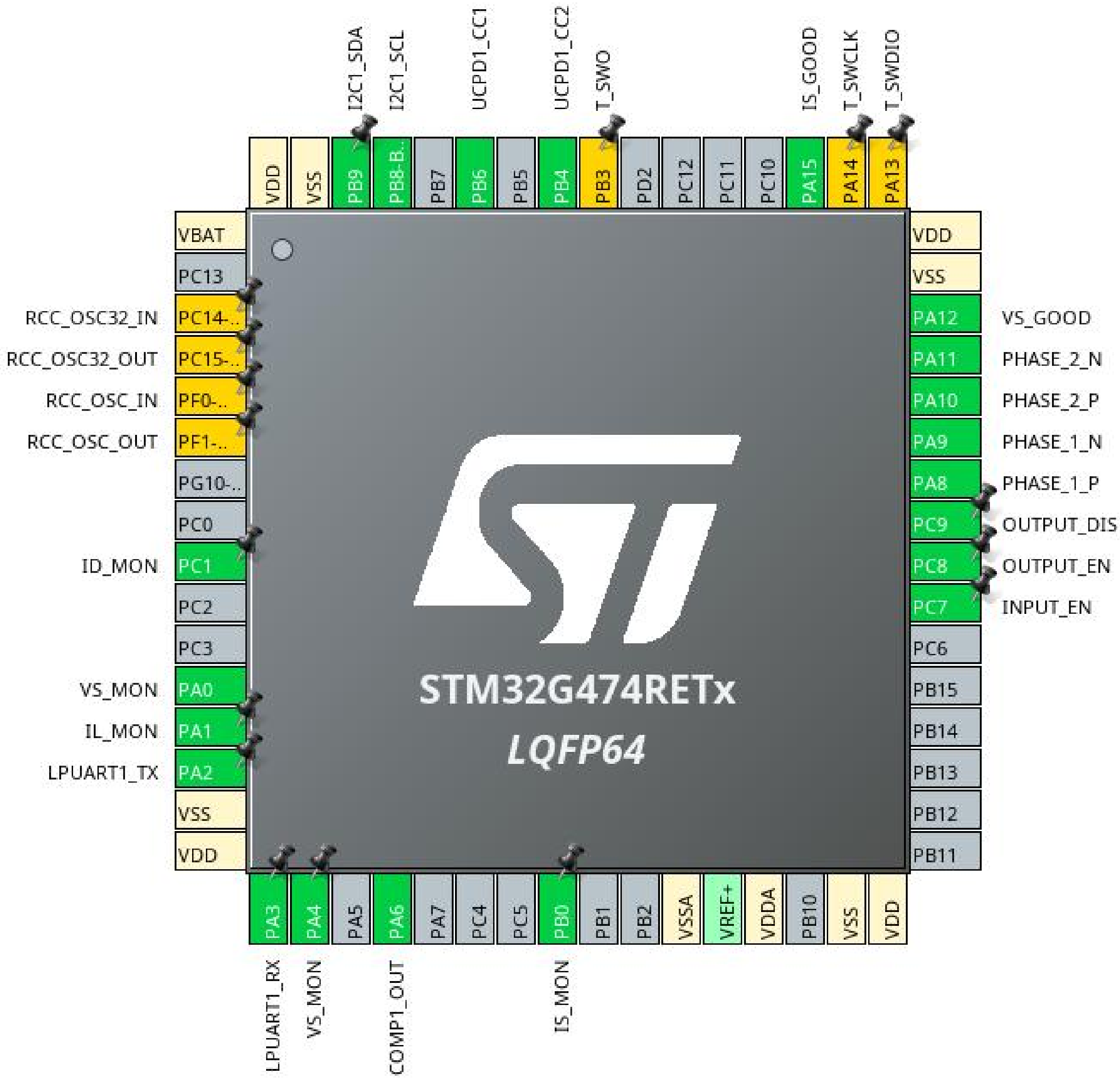
1.2. MCU

MCU Series	STM32G4
MCU Line	STM32G4x4
MCU name	STM32G474RETx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	ARM Cortex-M4

2. Pinout Configuration



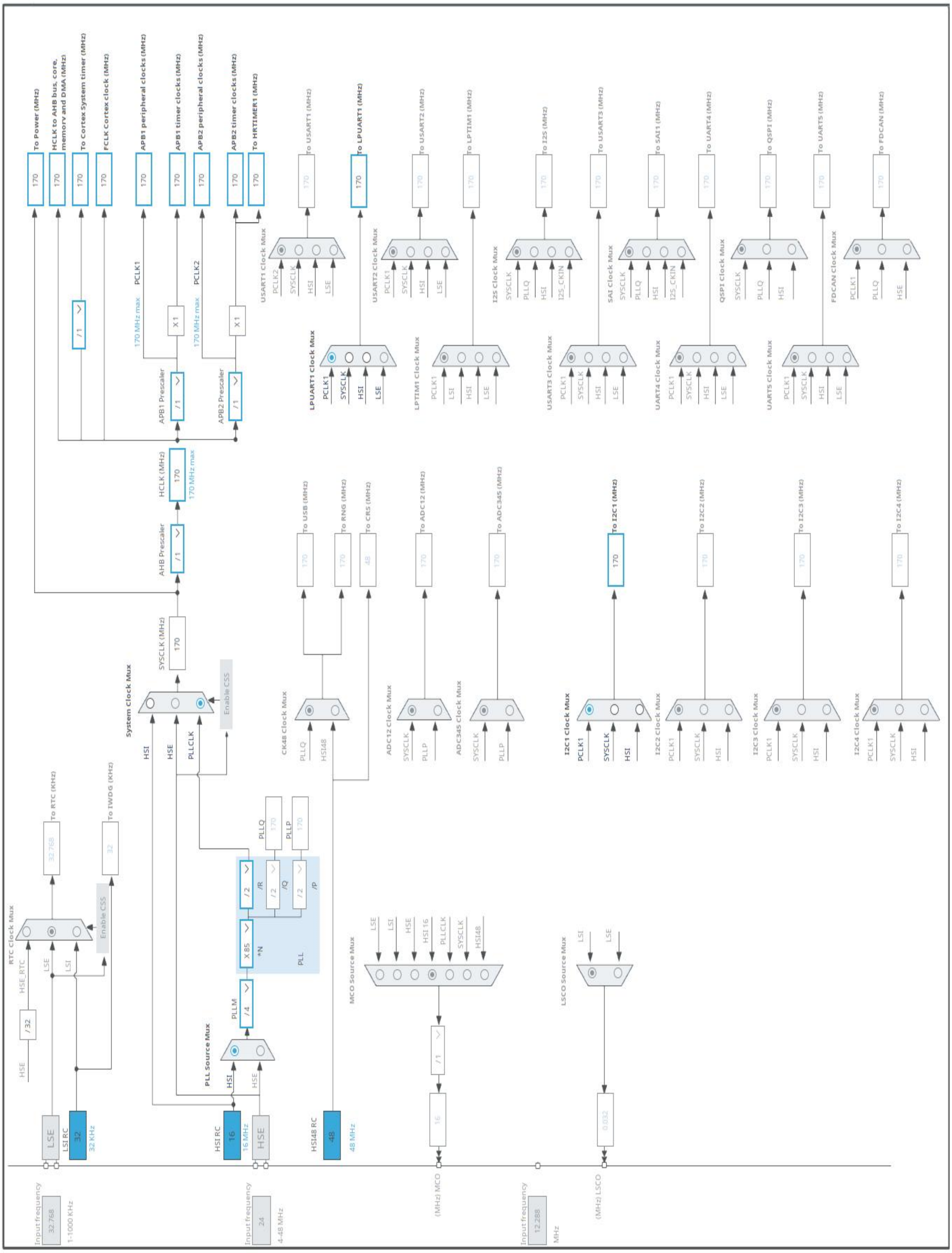
3. Pins Configuration

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
3	PC14-OSC32_IN *	I/O	RCC_OSC32_IN	RCC_OSC32_IN
4	PC15-OSC32_OUT *	I/O	RCC_OSC32_OUT	RCC_OSC32_OUT
5	PF0-OSC_IN *	I/O	RCC_OSC_IN	RCC_OSC_IN
6	PF1-OSC_OUT *	I/O	RCC_OSC_OUT	RCC_OSC_OUT
9	PC1	I/O	ADC1_IN7	ID_MON
12	PA0	I/O	ADC1_IN1	VS_MON
13	PA1	I/O	COMP1_INP, ADC1_IN2	IL_MON
14	PA2	I/O	LPUART1_TX	
15	VSS	Power		
16	VDD	Power		
17	PA3	I/O	LPUART1_RX	
18	PA4	I/O	ADC2_IN17	VS_MON
20	PA6	I/O	COMP1_OUT	
24	PB0	I/O	ADC1_IN15	IS_MON
27	VSSA	Power		
29	VDDA	Power		
31	VSS	Power		
32	VDD	Power		
39	PC7 **	I/O	GPIO_Output	INPUT_EN
40	PC8 **	I/O	GPIO_Output	OUTPUT_EN
41	PC9 **	I/O	GPIO_Output	OUTPUT_DIS
42	PA8	I/O	HRTIM1_CHA1	PHASE_1_P
43	PA9	I/O	HRTIM1_CHA2	PHASE_1_N
44	PA10	I/O	HRTIM1_CHB1	PHASE_2_P
45	PA11	I/O	HRTIM1_CHB2	PHASE_2_N
46	PA12	I/O	HRTIM1_FLT1	VS_GOOD
47	VSS	Power		
48	VDD	Power		
49	PA13 *	I/O	SYS_JTMS-SWDIO	T_SWDIO
50	PA14 *	I/O	SYS_JTCK-SWCLK	T_SWCLK
51	PA15	I/O	HRTIM1_FLT2	IS_GOOD
56	PB3 *	I/O	SYS_JTDO-SWO	T_SWO
57	PB4	I/O	UCPD1_CC2	
59	PB6	I/O	UCPD1_CC1	
61	PB8-BOOT0	I/O	I2C1_SCL	

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
62	PB9	I/O	I2C1_SDA	
63	VSS	Power		
64	VDD	Power		

** The pin is affected with an I/O function
 * The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32G4
Line	STM32G4x4
MCU	STM32G474RETx
Datasheet	DS12288_Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

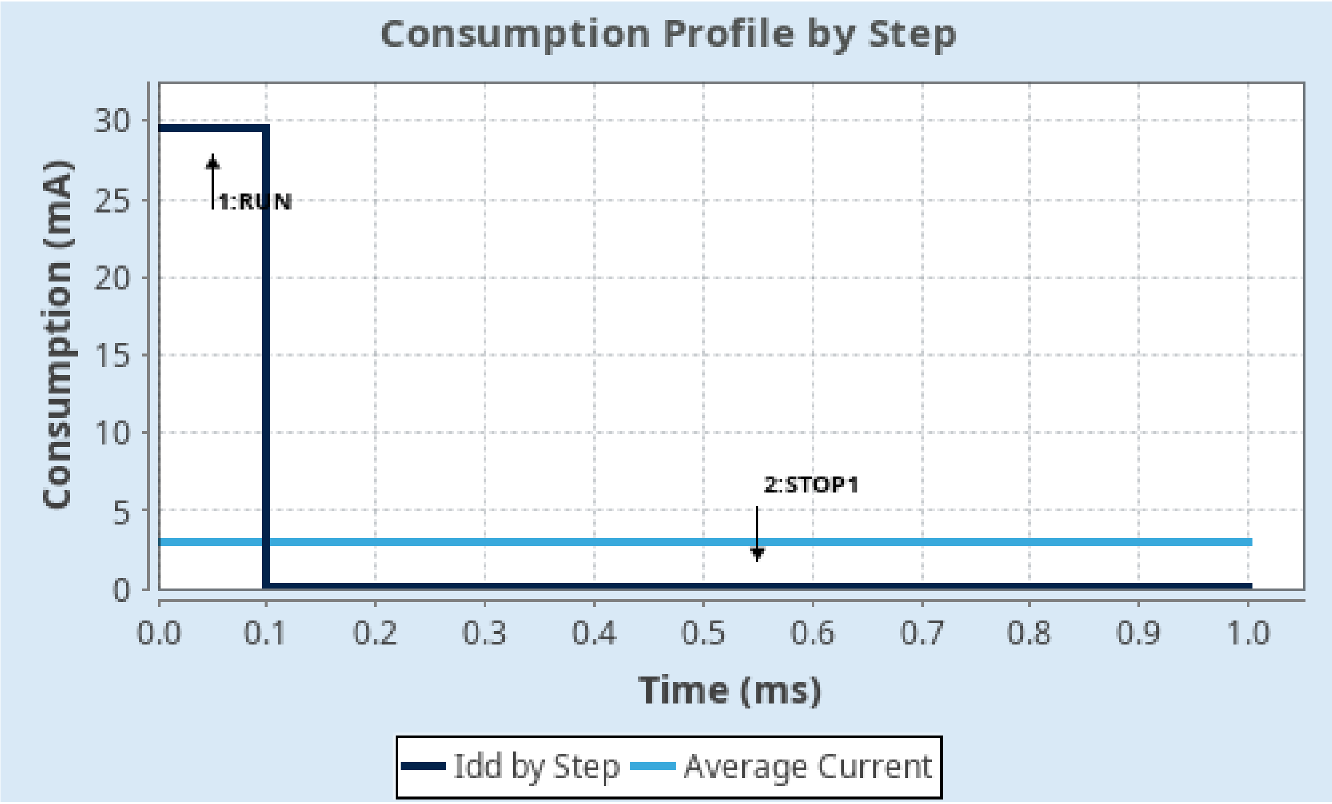
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP1
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-Boost	NoRange
Fetch Type	FLASH/DualBank/ART	NA
CPU Frequency	170 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL CLOCKS OFF
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	29.5 mA	80.5 µA
Duration	0.1 ms	0.9 ms
DMIPS	213.0	0.0
Ta Max	124.25	129.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	3.02 mA
Battery Life	1 month, 16 days, 9 hours	Average DMIPS	212.5 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
30
25
20
15
10
5
0
1:RUN
2:STOP1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Software Project

2.1. Project Settings

Name	Value
Project Name	final
Project Folder	/home/daniel/GitHub/PD_Charger/STM32CubeIDE/final
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_G4 V1.6.1
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0xC00
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO
2	MX_DMA_Init	DMA
3	MX_HRTIM1_Init	HRTIM1
4	SystemClock_Config	RCC
5	MX_UCPD1_Init	UCPD1
6	MX_USBPD_Init	USBPD
7	MX_LPUART1_UART_Init	LPUART1
8	MX_TRACER_EMB_Init	TRACER_EMB
9	MX_GUI_INTERFACE_Init	GUI_INTERFACE
10	MX_COMP1_Init	COMP1
11	MX_DAC3_Init	DAC3

Rank	Function Name	Peripheral Instance Name
12	MX_ADC2_Init	ADC2
13	MX_ADC1_Init	ADC1
14	MX_I2C1_Init	I2C1
15	MX_TCPP_Init	STMicroelectronics.X-CUBE-TCPP.4.2.0
16	MX_TCPP_Process	STMicroelectronics.X-CUBE-TCPP.4.2.0

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN1: IN1 Single-ended

IN2: IN2 Single-ended

IN7: IN7 Single-ended

mode: IN15

3.1.1. Parameter Settings:

ADCs_Common_Settings:	
Mode	Independent mode
ADC_Settings:	
Clock Prescaler	Synchronous clock mode divided by 4
Resolution	ADC 12-bit resolution
Data Alignment	Right alignment
Gain Compensation	0
Scan Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Low Power Auto Wait	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
Overrun behaviour	Overrun data preserved
ADC_Regular_ConversionMode:	
Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
Rank	1
Channel	Channel 1
Sampling Time	2.5 Cycles
Offset Number	No offset
ADC_Injected_ConversionMode:	
Enable Injected Conversions	Disable
Analog Watchdog 1:	
Enable Analog WatchDog1 Mode	false
Analog Watchdog 2:	
Enable Analog WatchDog2 Mode	false
Analog Watchdog 3:	

Enable Analog WatchDog3 Mode	false

3.2. ADC2
mode: IN17 Single-ended
3.2.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Independent mode

ADC_Settings:

Clock Prescaler	Synchronous clock mode divided by 4
Resolution	ADC 12-bit resolution
Data Alignment	Right alignment
Gain Compensation	0
Scan Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Low Power Auto Wait	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
Overrun behaviour	Overrun data preserved

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
Rank	1
Channel	Channel 17
Sampling Time	2.5 Cycles
Offset Number	No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

3.3. COMP1
mode: Input [+]
Input [-]: DAC3 OUT1
mode: ExternalOutput

3.3.1. Parameter Settings:

Basic Parameters:	
Trigger Mode	None
Hysteresis Level	None
Output Configuration:	
Blanking Source	None
Output Polarity	COMP output on GPIO isn't inverted

3.4. DAC3
mode: OUT1 mode
mode: OUT2 mode

3.4.1. Parameter Settings:

DAC Out1 Settings:	
Mode selected	Normal Mode
Output Buffer	Disable
DAC High Frequency	Mode Automatic
DMA Double Data	Disable
Signed Format	Disable
Trigger	None
Trigger2	None
User Trimming	Factory trimming
DAC Out2 Settings:	
Mode selected	Normal Mode
Output Buffer	Disable
DAC High Frequency	Mode Automatic
DMA Double Data	Disable
Signed Format	Disable
Trigger	None
Trigger2	None
User Trimming	Factory trimming

3.5. GUI_INTERFACE
mode: Enable

3.5.1. Parameter Settings:

Version	1.13.0
HWBoardVersionName	**PD_Charger ***
PDTypeName	**BUCKBOOST_SRC ***

3.6. HRTIM1
mode: Master Timer Enable
Timer A: TA1 and TA2 outputs active
Timer B: TB1 and TB2 outputs active
mode: Fault Input 1
mode: Fault Input 2

3.6.1. HRTIM Interrupt Configuration:

Sources:

1st Source of interrupt	No interrupt enabled
2nd Source of interrupt	No interrupt enabled
3rd Source of interrupt	No interrupt enabled
4th Source of interrupt	No interrupt enabled
5th Source of interrupt	No interrupt enabled
6th Source of interrupt	No interrupt enabled
7th Source of interrupt	No interrupt enabled
8th Source of interrupt	No interrupt enabled
9th Source of interrupt	No interrupt enabled

3.6.2. Synchro Configuration:

Master Timer Synchronization:

Sync Options	HRTIM instance doesn't handle external synchronization signals (SYNCIN, SYNCOUT)

3.6.3. High Resolution:

High Resolution Setting:

High Resolution to reach steps down to 184 ps (5.44GHz Equivalent Frequency for HRTIMclock=170MHz)	Enabled with Simple DLL Calibration

Calibration Settings:

Calibration Rate	Periodic DLL calibration: T : 2048 mult tHRTIM (0.012 ms)

Polling:

Timeout (in ms)	10

3.6.4. External Event Configuration:

External Event Clock:

Prescaler	fEEVS=fHRTIM

External Event 1:

Event Configuration	Disable

External Event 2:

Event Configuration	Disable

External Event 3:

Event Configuration	Disable

External Event 4:

Event Configuration	**Enable External Event 4 ***
Source	External Event Source 2 - COMP1
Event Polarity	External event is active high
Event Sensitivity	External event is active on level
Event Fast Mode	External Event is re-synchronized by the HRTIM logic before acting on outputs

External Event 5:

Event Configuration	Disable

External Event 6:

Event Configuration	Disable

External Event 7:

Event Configuration	Disable

External Event 8:

Event Configuration	Disable

External Event 9:

Event Configuration	Disable

External Event 10:

Event Configuration	Disable

3.6.5. Fault Lines Configuration:

Common Prescaler:

Prescaler	fFLTS=fHRTIM

Fault Line 1:

Line Configuration	Configure Fault Line
Source	Fault input is FLT input pin
Polarity	**External event is active low ***
Filter	Filter disabled
Lock	Fault settings bits are read/write
Line Ctl	Enable Fault Line
Counter on Faults (Threshold)	0
Reset Mode	Fault counter is reset on each reset / roll-over event
Blanking Source Type	Reset-Aligned Window - Blanking window starts on Timer A reset/roll-over and ends on Timer A COMP 3 event

Fault Line 2:

Line Configuration	Configure Fault Line
Source	Fault input is FLT input pin
Polarity	**External event is active low ***
Filter	Filter disabled
Lock	Fault settings bits are read/write
Line Ctl	Enable Fault Line
Counter on Faults (Threshold)	0
Reset Mode	Fault counter is reset on each reset / roll-over event
Blanking Source Type	Reset-Aligned Window - Blanking window starts on Timer B reset/roll-over and ends on Timer A COMP 3 event

Fault Line 3:

Line Configuration	No Configuration of Fault Line

Fault Line 4:

Line Configuration	No Configuration of Fault Line

Fault Line 5:

Line Configuration	No Configuration of Fault Line

Fault Line 6:

Line Configuration	No Configuration of Fault Line

3.6.6. ADC Triggers Configuration:

ADC Trigger 1:

ADC Trigger Configuration	Disable

ADC Trigger 2:

ADC Trigger Configuration	Disable

ADC Trigger 3:

ADC Trigger Configuration	Disable

ADC Trigger 4:

ADC Trigger Configuration	Disable

ADC Trigger 5:

ADC Trigger Configuration	Disable

ADC Trigger 6:

ADC Trigger Configuration	Disable

ADC Trigger 7:

ADC Trigger Configuration	Disable

ADC Trigger 8:

ADC Trigger Configuration	Disable

ADC Trigger 9:

ADC Trigger Configuration	Disable

ADC Trigger 10:

ADC Trigger Configuration	Disable

3.6.7. Burst Mode Configuration:

Burst Mode Enabling:

Burst Mode	Burst mode disabled

3.6.8. Master Timer:

General:

Timer Idx	Master Timer

Time Base Setting:

Prescaler Ratio	HRTIM Clock Multiplied by 32 (HRTIM Clock is set in Clock Configuration Tab with Max Value = 170MHz)
fHRCK Equivalent Frequency	5.44E9
Period	**0xFFDF ***
Resulting PWM Frequency	83050
Repetition Counter	**0x00 ***
Mode	The timer operates in continuous (free-running) mode

Timing Unit:

Interleaved Mode	Disabled
Start On Sync	Synchronization input event has no effect on the timer

Reset On Sync	Synchronization input event has no effect on the timer
Dac Synchro	No DAC synchronization event generated
Preload Enable	Preload disabled: the write access is directly done into the active register
Update Gating	Update done independently from the DMA burst transfer completion
Repetition Update	Update on repetition disabled
Burst Mode	Timer counter clock is maintained and the timer operates normally
Interrupt Requests Sources Selection : Please enter the number of Active Interrupt Requests	0
Number of Master Timer Internal DMA Request Sources - you first have to enable the Master Timer DMA Request in the DMA Settings Tab	0

Compare Unit 1:

Compare Unit 1 Configuration	Disable

Compare Unit 2:

Compare Unit 2 Configuration	Disable

Compare Unit 3:

Compare Unit 3 Configuration	Disable

Compare Unit 4:

Compare Unit 4 Configuration	Disable

Burst DMA Controller:

Burst DMA Configuration	Disable

3.6.9. Timer A:

General:

Timer Idx	Timer A
Basic/Advanced Configuration	Advanced (using HAL_Waveform methods)

Time Base Setting:

Prescaler Ratio	HRTIM Clock Multiplied by 32 (HRTIM Clock is set in Clock Configuration Tab with Max Value = 170MHz)
fHRCK Equivalent Frequency	5.44E9
Period	**0xFFDF ***
Resulting PWM Frequency	83050
Repetition Counter	**0x00 ***
Up Down Mode	Timer counter is operating in up-counting mode
Mode	The timer operates in continuous (free-running) mode

Timing Unit:

Interleaved Mode	Disabled
Start On Sync	Synchronization input event has no effect on the timer
Reset On Sync	Synchronization input event has no effect on the timer
Dac Synchro	No DAC synchronization event generated

Preload Enable	Preload disabled: the write access is directly done into the active register
Update Gating	Update done independently from the DMA burst transfer completion
Repetition Update	Update on repetition disabled
Burst Mode	Timer counter clock is maintained and the timer operates normally
Push Pull	Push-Pull mode disabled
Number of Faults to enable	0
Fault Lock	Timer fault enabling bits are read/write
Dead Time Insertion	**Deadtime is inserted between output 1 and output 2 ***
Delayed Protection Mode	No action
Update Trigger Sources Selection : Please enter the number of Triggers to select	0
Reset Update	Update by Timer reset / roll-over disabled
Resynchronized Update	Update taken into account immediately
Reset Trigger Sources Selection : Please enter the number of Triggers to select	0
Interrupt Requests Sources Selection : Please enter the number of Active Interrupt Requests	0
Number of Timer A Internal DMA Request Sources - you first have to enable the Timer A DMA Request in the DMA Settings Tab	0

Compare Unit 1:

Compare Unit 1 Configuration	Disable

Compare Unit 2:

Compare Unit 2 Configuration	Disable

Compare Unit 3:

Compare Unit 3 Configuration	Disable

Compare Unit 4:

Compare Unit 4 Configuration	Disable

Burst DMA Controller:

Burst DMA Configuration	Disable

Capture Unit 1:

Capture Unit 1 Configuration	Disable

Capture Unit 2:

Capture Unit 2 Configuration	Disable

External Event 1 Filtering:

Filtering Configuration	Disable

External Event 2 Filtering:

Filtering Configuration	Disable

External Event 3 Filtering:

Filtering Configuration	Disable

External Event 4 Filtering:

Filtering Configuration	Disable

External Event 5 Filtering:

Filtering Configuration	Disable

External Event 6 Filtering:

Filtering Configuration	Disable

External Event 7 Filtering:

Filtering Configuration	Disable

External Event 8 Filtering:

Filtering Configuration	Disable

External Event 9 Filtering:

Filtering Configuration	Disable

External Event 10 Filtering:

Filtering Configuration	Disable

External Event Counter A:

Event Counter A Configuration	Disable

Dead Time:

Dead Time Configuration	Enable
Prescaler (PSC - 16 bits value)	fDTG = fHRTIM * 8
Rising Value	**0x000 ***
Rising Sign	Positive deadtime on rising edge
Rising Lock	Deadtime rising value and sign is writable
Rising Sign Lock	Deadtime rising sign is writable
Falling Value	**0x000 ***
Falling Sign	Positive deadtime on falling edge
Falling Lock	Deadtime falling value and sign is writable
Falling Sign Lock	Deadtime falling sign is writable

Swap Output1 and Output2:

TA1 Output is sensitive to TA2 Control Registers and vice versa	Disable

Output 1 Configuration:

Output1 Configuration	TA1
Polarity	Output is active HIGH
Set Source Selection : Please enter the number of Active Set Sources	0
Reset Source Selection : Please enter the number of Active Reset Sources	0
Idle Mode	The output is not affected by the burst mode operation
Idle Level	Output at inactive level when in IDLE state
Fault Level	The output is not affected by the fault input
Chopper Mode Enable	Output signal is not altered
Burst Mode Entry Delayed	The programmed Idle state is applied immediately to the Output

Output 2 Configuration:

Output2 Configuration	TA2
Polarity	Output is active HIGH
Set Sources: nothing to set as Dead Time is enabled, Output2 signal is defined based on Output1 Configuration Set Sources	0
Reset Source Selection : Please enter the number of Active Reset Sources	0
Idle Mode	The output is not affected by the burst mode operation
Idle Level	Output at inactive level when in IDLE state
Fault Level	The output is not affected by the fault input
Chopper Mode Enable	Output signal is not altered
Burst Mode Entry Delayed	The programmed Idle state is applied immediately to the Output

Chopper Mode:

Chopper Mode Configuration	Disable

Dual DAC Channel Configuration:

Dual DAC Channel Trigger	Disabled

3.6.10. Timer B:

General:

Timer Idx	Timer B
Basic/Advanced Configuration	Advanced (using HAL_Waveform methods)

Time Base Setting:

Prescaler Ratio	HRTIM Clock Multiplied by 32 (HRTIM Clock is set in Clock Configuration Tab with Max Value = 170MHz)
fHRCK Equivalent Frequency	5.44E9
Period	**0xFFDF ***
Resulting PWM Frequency	83050
Repetition Counter	**0x00 ***
Up Down Mode	Timer counter is operating in up-counting mode
Mode	The timer operates in continuous (free-running) mode

Timing Unit:

Interleaved Mode	Disabled
Start On Sync	Synchronization input event has no effect on the timer
Reset On Sync	Synchronization input event has no effect on the timer
Dac Synchro	No DAC synchronization event generated
Preload Enable	Preload disabled: the write access is directly done into the active register
Update Gating	Update done independently from the DMA burst transfer completion
Repetition Update	Update on repetition disabled
Burst Mode	Timer counter clock is maintained and the timer operates normally
Push Pull	Push-Pull mode disabled
Number of Faults to enable	0

Fault Lock	Timer fault enabling bits are read/write
Dead Time Insertion	**Deadtime is inserted between output 1 and output 2 ***
Delayed Protection Mode	No action
Update Trigger Sources Selection : Please enter the number of Triggers to select	0
Reset Update	Update by Timer reset / roll-over disabled
Resynchronized Update	Update taken into account immediately
Reset Trigger Sources Selection : Please enter the number of Triggers to select	0
Interrupt Requests Sources Selection : Please enter the number of Active Interrupt Requests	0
Number of Timer B Internal DMA Request Sources - you first have to enable the Timer B DMA Request in the DMA Settings Tab	0

Compare Unit 1:

Compare Unit 1 Configuration	Disable

Compare Unit 2:

Compare Unit 2 Configuration	Disable

Compare Unit 3:

Compare Unit 3 Configuration	Disable

Compare Unit 4:

Compare Unit 4 Configuration	Disable

Burst DMA Controller:

Burst DMA Configuration	Disable

Capture Unit 1:

Capture Unit 1 Configuration	Disable

Capture Unit 2:

Capture Unit 2 Configuration	Disable

External Event 1 Filtering:

Filtering Configuration	Disable

External Event 2 Filtering:

Filtering Configuration	Disable

External Event 3 Filtering:

Filtering Configuration	Disable

External Event 4 Filtering:

Filtering Configuration	Disable

External Event 5 Filtering:

Filtering Configuration	Disable

External Event 6 Filtering:

Filtering Configuration	Disable

External Event 7 Filtering:

Filtering Configuration	Disable

External Event 8 Filtering:

Filtering Configuration	Disable

External Event 9 Filtering:

Filtering Configuration	Disable

External Event 10 Filtering:

Filtering Configuration	Disable

External Event Counter A:

Event Counter A Configuration	Disable

Dead Time:

Dead Time Configuration	Enable
Prescaler (PSC - 16 bits value)	fDTG = fHRTIM * 8
Rising Value	**0x000 ***
Rising Sign	Positive deadtime on rising edge
Rising Lock	Deadtime rising value and sign is writable
Rising Sign Lock	Deadtime rising sign is writable
Falling Value	**0x000 ***
Falling Sign	Positive deadtime on falling edge
Falling Lock	Deadtime falling value and sign is writable
Falling Sign Lock	Deadtime falling sign is writable

Swap Output1 and Output2:

TB1 Output is sensitive to TB2 Control Registers and vice versa	Disable

Output 1 Configuration:

Output1 Configuration	TB1
Polarity	Output is active HIGH
Set Source Selection : Please enter the number of Active Set Sources	0
Reset Source Selection : Please enter the number of Active Reset Sources	0
Idle Mode	The output is not affected by the burst mode operation
Idle Level	Output at inactive level when in IDLE state
Fault Level	The output is not affected by the fault input
Chopper Mode Enable	Output signal is not altered
Burst Mode Entry Delayed	The programmed Idle state is applied immediately to the Output

Output 2 Configuration:

Output2 Configuration	TB2
Polarity	Output is active HIGH
Set Sources: nothing to set as Dead Time is enabled, Output2 signal is defined based on Output1 Configuration Set Sources	0
Reset Source Selection : Please enter the number of	0

Active Reset Sources	
Idle Mode	The output is not affected by the burst mode operation
Idle Level	Output at inactive level when in IDLE state
Fault Level	The output is not affected by the fault input
Chopper Mode Enable	Output signal is not altered
Burst Mode Entry Delayed	The programmed Idle state is applied immediately to the Output

Chopper Mode:

Chopper Mode Configuration	Disable

Dual DAC Channel Configuration:

Dual DAC Channel Trigger	Disabled

3.7. I2C1

I2C: I2C

3.7.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	100
Fall Time (ns)	100
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	**0x40B285C2 ***

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.8. LPUART1

Mode: Asynchronous

3.8.1. Parameter Settings:

Basic Parameters:

Baud Rate	921600

Word Length	**7 Bits (including Parity) ***
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.9. RCC

3.9.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Disabled
Data Cache	Enabled
Flash Latency(WS)	4 WS (5 CPU cycle)

RCC Parameters:

HSI Calibration Value	64
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1 boost

Peripherals Clock Configuration:

Generate the peripherals clock configuration	TRUE

3.10. SYS

Timebase Source: SysTick

mode: save power of non-active UCPD - deactive Dead Battery pull-up

3.11. TRACER_EMB

Uart Trace Source: LPUART1

3.11.1. Parameter Settings:

Version	TRACER_EMB
TRACER_EMB request LPUART1 TX DMA	enabled
TRACER_EMB request LPUART1 NVIC	enabled

3.12. UCPD1

UCPD Mode: Source

3.12.1. Parameter Settings:

Version	1.0

3.13. FREERTOS

Interface: CMSIS_V1

3.13.1. Config parameters:

API:	
FreeRTOS API	CMSIS v1
Versions:	
FreeRTOS version	10.3.1
CMSIS-RTOS version	1.02
MPU/FPU:	
ENABLE_MPU	Disabled
ENABLE_FPU	Disabled
Kernel settings:	
USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000
MAX_PRIORITIES	7
MINIMAL_STACK_SIZE	128

MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Disabled
USE_COUNTING_SEMAPHORES	Disabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Enabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled

Memory management settings:

Memory Allocation	Dynamic
TOTAL_HEAP_SIZE	**7000 ***
Memory Management scheme	heap_4

Hook function related definitions:

USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Disabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Disabled

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

3.13.2. Include parameters:

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Disabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Disabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Disabled
xTaskGetCurrentTaskHandle	Disabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Disabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.13.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Disabled

Project settings (see parameter description first):

Use FW pack heap file	Enabled

3.14. STMicroelectronics.X-CUBE-TCPP.4.2.0
mode: DeviceJjUSBPDOoApplication
mode: BoardOoPartJjtcpp0203
mode: BoardOoSupportJjXAaNUCLEOAaSRC1M1

3.15. USBPD
mode: Port Configuration

Stack Configuration: Full Stack
Timer service Source: TIM1
mode: Tracer Source (TRACER_EMB)

3.15.1. Parameter Settings:

USBPD needs:

USBPD request UCPD1 NVIC	enabled
USBPD request UCPD1 DMA	enabled

3.15.2. DPM Core Parameters:

USB IF and Manufacturer ID:

Vendor ID	**0x0483 ***
Product ID	0x0002
XID	0xF0000003

3.15.3. PDO Sources:

Number of PDO Source:

Number of PDO to define	**2 ***

Generic Source Parameters:

Unchunked Extended Messages Supported	Not supported
Dual-Role Data	Supported
USB Communication Capable	Not supported
Unconstrained Power	Not supported
USB Suspend Supported	Not supported
Dual Role Power	Not supported

PDO 0:

PDO type	Fixed Supply (Vmin=Vmax)
Voltage (mV)	5000
Current (mA)	100
Peak Current	Peak equal

PDO 1:

PDO type	Fixed Supply (Vmin=Vmax)
Voltage (mV)	5000
Current (mA)	**1500 ***
Peak Current	Peak equal

3.15.4. Stack Port 0 Parameters:

Port Configuration:	
UCPD Instance	UCPD1
DMA Request RX for UCPD Port 0	UCPD1_RX_DMA1_Channel_4
DMA Request TX for UCPD Port 0	UCPD1_TX_DMA1_Channel_2
Start of Packet Parameters:	
SOP	Supported
SOP'	Not supported
SOP"	Not supported
SOP' debug	Not supported
SOP" debug	Not supported
Port 0 Parameters:	
Specification revision value	Revision 3 (PD3)
Default port role	Source
Port role swap	Not supported
Data role swap to DFP	Supported
Data role swap to UFP	Supported
Vendor defined messages	Not supported
Discover Identity response	Not supported
Discover Identity sent	Not supported
Ping	Not supported
Caps counter	Not supported
PD Revision 3 specific parameters:	
Unchunk mode	Not supported
Fast role swap	Not supported
Higher Capability	**Supported ***
USB Communication Capable	Not supported
Unconstrained Power	Not supported
USB Suspend Supported	Not supported
PPS message	Not supported
Source Capabilities Extended message	**Supported ***
Alert message	Not supported
Status message	Not supported
Manufacturer Info message	Not supported
Country Codes message	Not supported
Country Info message	Not supported
Security Response message	Not supported
Firmware update Response message	Not supported
Get Battery Capabitity and Status messages	Not supported
Cable Detection Parameters:	

CAD default resistor	Default USB Power
CAD accessory	Not supported

3.15.5. PDO General Definitions:

Number of Source PDOs for port 0	1

3.15.6. User Port 0 Parameters:

Port 0 Parameters:

Data role swap	Not supported
VCONN swap	Not supported

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PC1	ADC1_IN7	Analog mode	No pull-up and no pull-down	n/a	ID_MON
	PA0	ADC1_IN1	Analog mode	No pull-up and no pull-down	n/a	VS_MON
	PA1	ADC1_IN2	Analog mode	No pull-up and no pull-down	n/a	IL_MON
	PB0	ADC1_IN15	Analog mode	No pull-up and no pull-down	n/a	IS_MON
ADC2	PA4	ADC2_IN17	Analog mode	No pull-up and no pull-down	n/a	VS_MON
COMP1	PA1	COMP1_INP	Analog mode	No pull-up and no pull-down	n/a	IL_MON
	PA6	COMP1_OUT	Alternate Function Push Pull	No pull-up and no pull-down	Low	
HRTIM1	PA8	HRTIM1_CHA1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PHASE_1_P
	PA9	HRTIM1_CHA2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PHASE_1_N
	PA10	HRTIM1_CHB1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PHASE_2_P
	PA11	HRTIM1_CHB2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PHASE_2_N
	PA12	HRTIM1_FLT1	Alternate Function Push Pull	No pull-up and no pull-down	Low	VS_GOOD
	PA15	HRTIM1_FLT2	Alternate Function Push Pull	No pull-up and no pull-down	Low	IS_GOOD
HRTIM1	PA8	HRTIM1_CHA1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PHASE_1_P
	PA9	HRTIM1_CHA2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PHASE_1_N
	PA10	HRTIM1_CHB1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PHASE_2_P
	PA11	HRTIM1_CHB2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PHASE_2_N
	PA12	HRTIM1_FLT1	Alternate Function Push Pull	No pull-up and no pull-down	Low	VS_GOOD
	PA15	HRTIM1_FLT2	Alternate Function Push Pull	No pull-up and no pull-down	Low	IS_GOOD
I2C1	PB8-BOOT0	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PB9	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
LPUART1	PA2	LPUART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA3	LPUART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
UCPD1	PB4	UCPD1_CC2	Analog mode	No pull-up and no pull-down	n/a	
	PB6	UCPD1_CC1	Analog mode	No pull-up and no pull-down	n/a	
Single Mapped Signals	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a	RCC_OSC32_IN
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	RCC_OSC32_OUT
	PF0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	RCC_OSC_IN
	PF1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	RCC_OSC_OUT
	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	T_SWDIO
	PA14	SYS_JTCK-	n/a	n/a	n/a	T_SWCLK

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
		SWCLK				
	PB3	SYS_JTDO-SWO	n/a	n/a	n/a	T_SWO
GPIO	PC7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	INPUT_EN
	PC8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	OUTPUT_EN
	PC9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	OUTPUT_DIS

4.2. DMA configuration

DMA request	Stream	Direction	Priority
UCPD1_TX	DMA1_Channel2	Memory To Peripheral	Low
UCPD1_RX	DMA1_Channel4	Peripheral To Memory	Low
LPUART1_TX	DMA1_Channel3	Memory To Peripheral	Low
ADC1	DMA1_Channel5	Peripheral To Memory	Low
ADC2	DMA1_Channel6	Peripheral To Memory	Low

UCPD1_TX: DMA1_Channel2 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

UCPD1_RX: DMA1_Channel4 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

LPUART1_TX: DMA1_Channel3 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

ADC1: DMA1_Channel5 DMA request Settings:

Mode:	**Circular ***
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Half Word

Memory Data Width:	Half Word

ADC2: DMA1_Channel6 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Half Word
Memory Data Width:	Half Word

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
DMA1 channel2 global interrupt	true	5	0
DMA1 channel3 global interrupt	true	3	0
DMA1 channel4 global interrupt	true	5	0
DMA1 channel5 global interrupt	true	3	0
DMA1 channel6 global interrupt	true	3	0
UCPD1 interrupt / UCPD1 wake-up interrupt through EXTI line 43	true	5	0
LPUART1 global interrupt	true	6	0
PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/38/39/40/41	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC1 and ADC2 global interrupt	unused		
I2C1 event interrupt / I2C1 wake-up interrupt through EXTI line 23	unused		
I2C1 error interrupt	unused		
TIM6 global interrupt, DAC1 and DAC3 channel underrun error interrupts	unused		
COMP1, COMP2 and COMP3 interrupts through EXTI lines 21, 22 and 29	unused		
HRTIM master timer global interrupt	unused		
HRTIM timer A global interrupt	unused		
HRTIM timer B global interrupt	unused		
HRTIM fault global interrupt	unused		
FPU global interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler

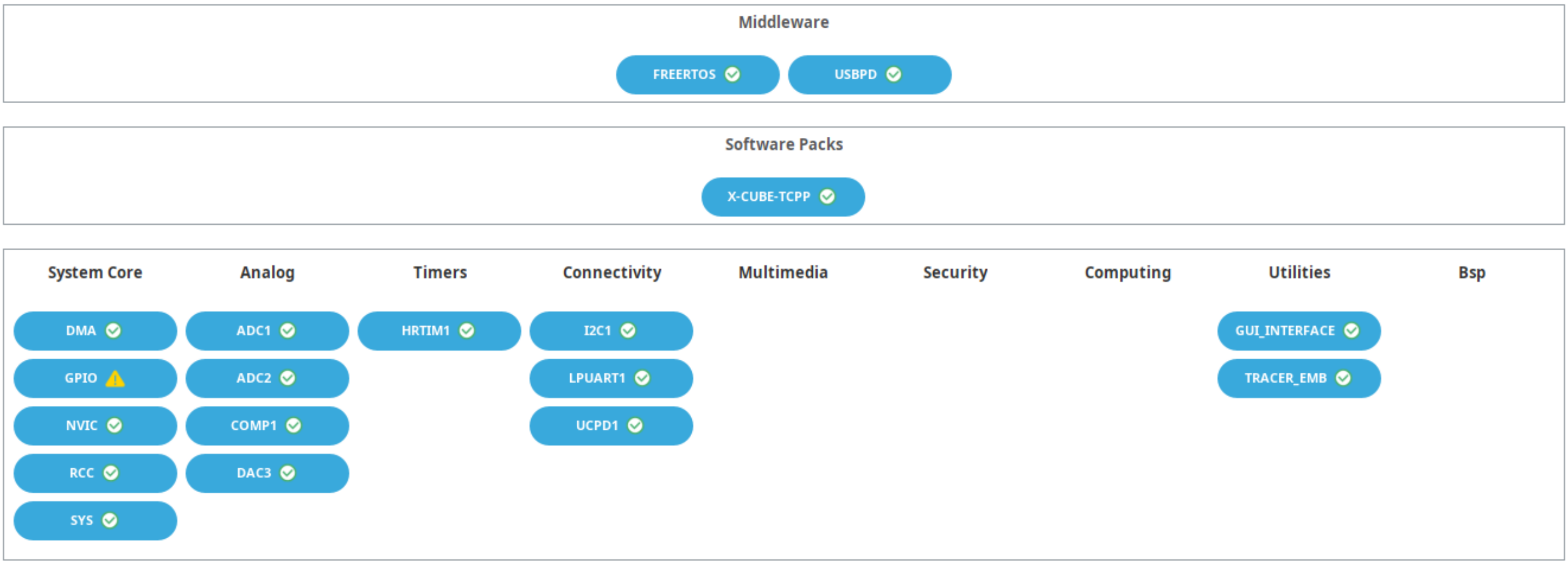
Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	true	true
DMA1 channel2 global interrupt	false	true	true
DMA1 channel3 global interrupt	false	true	true
DMA1 channel4 global interrupt	false	true	true
DMA1 channel5 global interrupt	false	true	true
DMA1 channel6 global interrupt	false	true	true
UCPD1 interrupt / UCPD1 wake-up interrupt through EXTI line 43	false	true	false
LPUART1 global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current



6. Software Pack Report

6.1. Software Pack selected

Vendor	Name	Version	Component
STMicroelectronics	X-CUBE-TCPP	4.2.0	Class : Device Group : Application Variant : Source Version : 4.2.0 Class : Board Part Group : tcpp0203 SubGroup : tcpp0203 Version : 1.2.3 Class : Board Support Group : X-NUCLEO-SRC1M1 SubGroup : Common Version : 1.2.1

7. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32g4_bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32g4_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32g4_svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-usb-c-pd-solutions-presentation.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32g4-series-product-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf
Brochures	https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32g4.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flpowerstbd.pdf
Flyers	https://www.st.com/resource/en/flyer/fldpstpfc11120.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-

	stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4232-getting-started-with-analog-comparators-for-stm32f3-series-and-stm32g4-series-devices-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4296-use-stm32f3stm32g4-ccm-sram-with-iar-embedded-workbench-keil-mdkarm-stmicroelectronics-stm32cubeide-and-other-gnubased-toolchains-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5093-getting-started-with-stm32g4-series--hardware-development-boards-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5306-operational-amplifier-opamp-usage-in-stm32g4-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5310-guideline-for-using-analog-features-of-stm32g4-series-versus-stm32f3-series-devices-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5315-stm32cube-firmware-examples-for-stm32g4-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5346-stm32g4-adc-use-tips-and-recommendations-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5094-migrating-between-stm32f334303-lines-and-stm32g431xxg474xxg491xx-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5738-stm32g4-series-lifetime-estimates-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4760-quadspi-interface-on-stm32-microcontrollers-and-microprocessors--stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5788-stm32-digital-power-pid-and-iir-filters-for-smps-control-design-and-comparison-on-bg414edpow1-discovery-kit-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4991-how-to-wake-

	up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5325-how-to-use-the-cordic-to-perform-mathematical-functions-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4894-how-to-use-eeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signaltonoise-ratio-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5978-introduction-to-mb1971-llc-hat-12-v-to-75-v1-a-for-f334-g474-nucleo-board-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2548-introduction-to-dma-controller-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4277-how-to-use-

	pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-with-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4908-getting-started-with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5224-introduction-to-dmamux-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5348-introduction-to-fdcan-peripherals-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4566-how-to-extend-the-dac-performance-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5647-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2606-introduction-to-system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4435-guidelines-for-

for related Tools & Software	obtaining-ulcsaiec-607301603351-class-b-certification-in-any-stm32-application-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4657-stm32-inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4841-digital-signal-processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5056-integration-guide-for-the-xcubesbsfu-stm32cube-expansion-package-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5305-digital-filter-implementation-with-the-fmac-using-stm32cubeg4-mcu-package-stmicroelectronics.pdf
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